Edge-Cell Graph Encoding and Fixed-Point Arithmetic for FPGA Growing Neural Gas

Integer-only online topology learning on a sub-10K-LUT FPGA

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-41%
total graph-state memory
2,000 B -> 1,180 B

~625 us
deterministic latency
~16.9k cycles @ 27 MHz

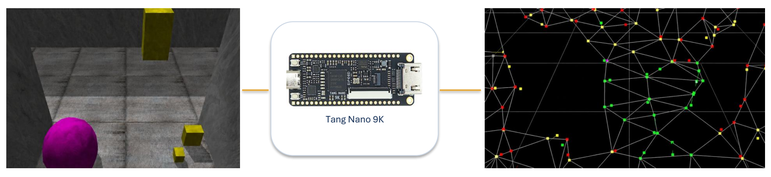
38%
LUT utilization
3,205 / 8,640 LUTs

~1,600/s
online throughput
UART-measured FPGA snapshots

1 · Problem & Motivation

Edge robots need on-board unsupervised adaptation.

- Streaming raw sensor data to the cloud is often impractical under intermittent links and latency limits.
- GNG learns a topology incrementally, but dynamic edges and floating point are expensive on tiny FPGAs.
- The target is bounded memory, deterministic per-sample time, and an integer-only datapath.



Unstructured field scene -> integer FPGA engine -> online topology graph

2 · Contributions in the Paper

- Synthesized GNG engine on Gowin GW1NR-9C with no CPU, no FPU, and no PLL.
- Edge-cell graph storage: upper triangle only, one 8-bit age+1 cell per possible undirected edge.
- Q8/Q16 shift-scaled learning rates mapped to DSP-friendly multiply-and-shift updates.
- Winner-indexed fused NB_SCAN combines error decay, edge aging, neighbor movement, and pruning.

No soft-core CPU
FSM datapath

No FPU
Q8/Q16 shifts

No dynamic memory
fixed BRAM arrays

3 · FPGA Architecture

Single-clock FSM at 27 MHz with BRAM-mapped state.

UART input
100-sample dataset ROM

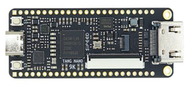
WIN_SCAN
nearest s1, runner-up s2

UPDATE
winner error + Q8 move

NB_SCAN
node BRAM + edge BRAM RMW

CONNECT / INSERT
degree counters + split edge

DBG/SNAP
cycle counter + topology stream



Node, edge, and dataset BRAMs use synchronous read-modify-write access; each sequence costs at least 3 cycles.

4 · GNG Loop and Fused NB_SCAN

Paper-aligned FSM sequence per input sample

SAMPLE	fetch input xi
WIN_SCAN	find s1 and s2
UPDATE	err_s1 += d2 >> ERR_SHIFT; move s1 by EPS_WIN_Q8
NB_SCAN	decay err_n; probe edge_cell[idx(s1,n)]
ACTIVE EDGE	age/prune edge; move neighbor by EPS_N_Q16
CONNECT	reset edge(s1,s2) age; update degree counters
INSERT / SNAP	every lambda split max-error edge; stream UART snapshots

Why fused?

- Canonical GNG separates global error decay from winner-neighbor edge updates.
- The FPGA already scans all node slots around s1, so decay is folded into the same NB_SCAN.
- Per-node degree counters detect isolated nodes without a second full edge-array scan.

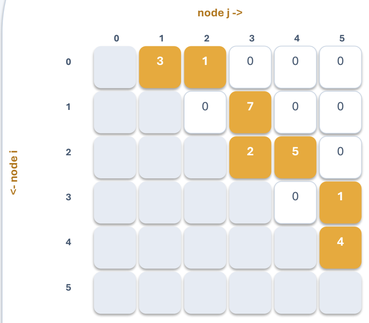
Hardware invariant

Only fixed-capacity arrays are used: node state is stored once, edge state is stored once, and dynamic topology is expressed by age+1 cells and degree counters. This keeps memory bounded even as GNG rewires.

State	BRAM word	Role
node	x,y,deg,err	coordinates + degree + error
edge	age+1	0 = inactive; v>0 = age v-1
dataset	x,y sample	100-entry on-chip input set

5 · Edge-Cell Encoding

Flatten only the upper triangle of an undirected graph.



cell = age + 1

0 means no edge

v > 0 means true age = v - 1

Half-adjacency indexing, row by row

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0,1	0,2	0,3	0,4	0,5	1,2	1,3	1,4	1,5	2,3	2,4	2,5	3,4	3,5	4,5

idx(i,j) = i(2Nmax - i - 1)/2 + (j - i - 1)

(1,3) -> cell 6

Example uses Nmax = 6: 1(12 - 1 - 1)/2 + (3 - 1 - 1) = 5 + 1 = 6

Dense matrix
40 x 40 x 8-bit
= 1,600 B

->

Edge-cell array
40 x 39 / 2 bytes
= 780 B

51.25% less edge storage; total graph state drops 2,000 B -> 1,180 B.

This matches the paper's fixed-capacity allocation: graph growth and rewiring do not allocate memory dynamically; they only update existing node and edge BRAM cells.

Presence + age
one byte encodes both

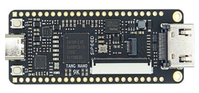
Deterministic address
13-bit BRAM index at Nmax=40

Degree counters
isolated nodes need no edge scan

6 · Fixed-Point Arithmetic

Integer-only datapath

Coordinates are 16-bit signed integers in [-1000, 1000] after min-max normalization.



Coordinate representation

x_norm = ((x - x_min) / (x_max - x_min)) * 2S - S, S = 1000

80-bit node word

x [16]	y [16]	deg [8]	err [32]

Squared-distance unit

dx = xi_x - x_n, dy = xi_y - y_n
d2 = dx^2 + dy^2 // 17-bit signed -> 35-bit unsigned
err_s1 += d2 >> ERR_SHIFT

Winner adaptation, Q8

Delta x_s1 = floor(EPS_WIN_Q8 * (xi_x - x_s1) / 2^8)
Delta y_s1 = floor(EPS_WIN_Q8 * (xi_y - y_s1) / 2^8)

Neighbor adaptation, Q16

Delta x_n = floor(EPS_N_Q16 * (xi_x - x_n) / 2^16)
Delta y_n = floor(EPS_N_Q16 * (xi_y - y_n) / 2^16)

Multiplierless error decay

err_n = err_n - (err_n >> ERR_DECAY_SHIFT)
beta approx 1 - 2^-8 = 0.9961

Parameter	Value	Hardware mapping
Nmax	40	node BRAM depth
Amax	50	A_MAX_STORED = 51
lambda	100	lambda counter
eps_w	~0.301	EPS_WIN_Q8 = 77, >> 8
eps_n	~0.001	EPS_N_Q16 = 66, >> 16
alpha	0.5	INS_ALPHA_SHIFT = 1
beta	~0.9961	ERR_DECAY_SHIFT = 8

Where 77 and 66 come from

EPS_Q = round(epsilon * 2^Q). The implementation targets a winner rate near 0.3 and a neighbor rate near 0.001, so EPS_WIN_Q8 = round(0.301 * 256) = 77 and EPS_N_Q16 = round(0.001 * 65536) = 66.

Interpretation

The fixed-point approximation is not exact, but the paper reports low final QE/TE on both datasets, so the hardware-friendly rates are acceptable for the evaluated benchmarks.

7 · Fits on a 9K FPGA

Place-and-route on Gowin GW1NR-LV9QN88PC6

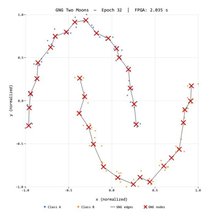
Resource	Used	Avail.	Util.
LUT	3,205	8,640	38%
FF	1,408	6,693	22%
BSRAM	6	26	24%
DSP	5	10	50%

Memory	Baseline	Proposed	Saving
Node array	400 B	400 B	-
Edge memory	1,600 B	780 B	-51.25%
Total graph	2,000 B	1,180 B	-41.0%

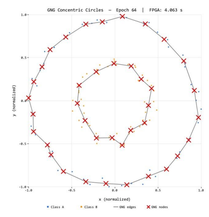
The design retains 62% LUT and 76% BRAM headroom for application logic.

8 · Learning Quality

Final FPGA snapshots preserve the target topology.



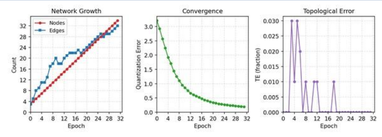
Two Moons, epoch 32
QE 0.1832, TE 0.00



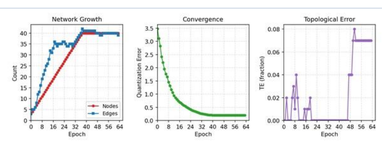
Concentric Circles, epoch 64
QE 0.1960, TE 0.07

The paper reports QE decreasing on both datasets. TE stays near zero during growth; Circles drifts after saturation because the fixed Nmax graph cannot insert more nodes across the inter-ring gap.

Dataset	Epoch	Nodes	Edges	QE	TE
Two Moons	32	35	33	0.1832	0.00
Two Moons	64	40	34	0.1455	0.01
Circles	32	35	36	0.2592	0.00
Circles	64	40	39	0.1960	0.07



Two Moons training dynamics
node/edge growth, QE convergence, TE

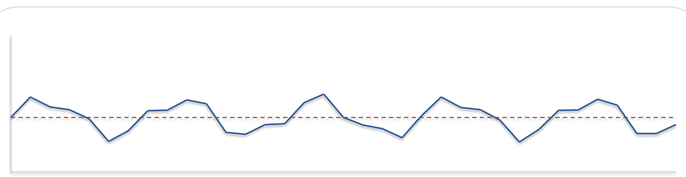


Concentric Circles training dynamics
64 epochs after Nmax saturation

9 · Deterministic Real-Time Throughput

Measured directly from the FPGA cycle counter embedded in DBG frames.

Dataset	Cycles	us	Samples/s	Total s
Two Moons	16,852	624.1	1,602	5.40
Circles	16,931	627.1	1,595	5.72



~1,600 samples/s with bounded per-sample latency

10 · Takeaway and Future Work

Self-organizing clustering with dynamic topology is feasible on a sub-10K-LUT FPGA when memory layout and arithmetic format are co-designed.

- Current evaluation: static 2D benchmarks; streaming non-stationary data remains future work.
- Next: parallel winner-search tree to reduce O(Nmax) latency toward O(log Nmax).
- Next: configurable Q-format constants and multi-stream sensor input.
- Target deployment: continual topology learning on low-power, memory-constrained robots/devices.

Code, poster, slides, and data

github.com/tzf230201/fpga_gng

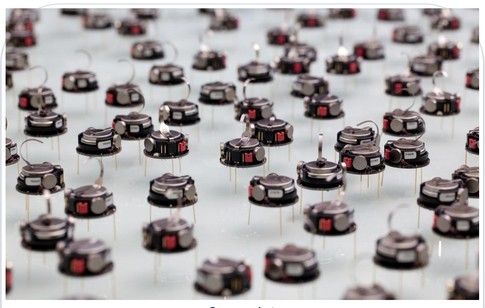
Prospective Edge-Robotics Applications



Lightweight flapping robot
payload-limited topology mapping



Bio-inspired robot
strict body and actuator constraints



Swarm robot
distributed intelligence, low memory



Long-term exploration robot
low power, adaptive autonomy



Infrastructure inspection
narrow spaces, local computation

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